

Year 7 · Chapter 1 · Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

Step 1: What is an integer?

E1 (WE DO) Where is the dot?
the dot is at -2

1 Where is the dot?
the dot is at -4

2 Where is the dot?
the dot is at 3

3 Where is the dot?
the dot is at -7

4 Where is the dot?
the dot is at -1

5 Which integer is 6 left of 0?
it is at -6

Step 2: Integers in real life

E2 (WE DO) 3°C below zero
 $= -3$

6 a gain of 4 kg
 $= 4$

7 8 m below sea level
 $= -8$

8 a loss of \$12
 $= -12$

9 6 floors down
 $= -6$

10 a rise of 15 cm
 $= 15$

11 spending \$30
 $= -30$

Step 3: Opposites

E3 (WE DO) 7
opposite: -7

12 -5
opposite: 5

13 9
opposite: -9

14 -12
opposite: 12

15 30
opposite: -30

16 0
opposite: 0

17 -100
opposite: 100

Step 4: Double opposites

E4 (WE DO) $-(-3)$
 $= \text{opposite of } -3 \cdot = 3$

E5 (WE DO) $-(+4)$
 $= \text{opposite of } 4 \cdot = -4$

18 $-(-8)$
 $= \text{opposite of } -8 \cdot = 8$

19 $-(+5)$
 $= \text{opposite of } 5 \cdot = -5$

20 $-(-20)$
 $= \text{opposite of } -20 \cdot = 20$

21 $-(+12)$
 $= \text{opposite of } 12 \cdot = -12$

22 $-(-45)$
 $= \text{opposite of } -45 \cdot = 45$

23 $-(+30)$
 $= \text{opposite of } 30 \cdot = -30$

Step 5: Distance from zero

E6 (WE DO) -4 or 3
distances: 4 and 3 $\cdot$ further: -4

24 -8 or 5
distances: 8 and 5 $\cdot$ further: -8

25 6 or -2
distances: 6 and 2 $\cdot$ further: 6

26 -10 or 9
distances: 10 and 9 $\cdot$ further: -10

27 -3 or -11
distances: 3 and 11 $\cdot$ further: -11

28 14 or -15
distances: 14 and 15 $\cdot$ further: -15

Step 6: Is it right?

E7 (WE DO) Tom says $-(-4) = -4$. Is he right?
 $= \text{opposite of } -4 \cdot = 4 \cdot$ No. The answer is 4.

29 Ali says $-(+7) = 7$. Is he right?
 $= \text{opposite of } 7 \cdot = -7 \cdot$ No. The answer is -7 .

30 Ben says -6 is further from 0 than 4. Is he right?
distances: 6 and 4 $\cdot$ 6 is more than 4 $\cdot$ Yes. Ben is right.

31 Mia says the opposite of 0 is 0. Is she right?
0 is not left or right of 0 $\cdot$ so its opposite is itself $\cdot$ Yes. Mia is right.

32 Jo says 10 m down and 10 m up are opposites. Is Jo right?
down: -10 , up: 10 $\cdot$ same distance, other side $\cdot$ Yes. Jo is right.

1.02 The number line (booklet p. 5)

Step 1: Reading a number line

E1 (WE DO) Where is P?
P is at -5

1 Where is Q?

Q is at -1

2 Where is R?

R is at 3

3 Where is S?

S is at -7

4 Where is T?

T is at 5

5 Which integer is 3 right of -5?

land on -2

Step 2: Moving left and right

E2 (WE DO) Start at 3. Move 5 left.

land on -2

6 Start at -2. Move 6 right.

land on 4

7 Start at 1. Move 4 left.

land on -3

8 Start at -3. Move 5 left.

land on -8

9 Start at -7. Move 10 right.

land on 3

10 Start at 6. Move 13 left.

land on -7

Step 3: Distance between

E3 (WE DO) -2 and 4

2 to 0, 4 from 0 · = 6

E4 (WE DO) -7 and -3

count -7 to -3 · = 4

11 -1 and 6

1 to 0, 6 from 0 · = 7

12 -5 and 3

5 to 0, 3 from 0 · = 8

13 -9 and -2

count -9 to -2 · = 7

14 4 and -8

8 to 0, 4 from 0 · = 12

15 -12 and -5

count -12 to -5 · = 7

16 -15 and 10

15 to 0, 10 from 0 · = 25

Step 4: Scales in 2s, 5s and 10s

E5 (WE DO) What is at the ?

$10 \div 5 = 2$, step 2 · ? = 8

17 What is at the ?

$8 \div 4 = 2$, step 2 · ? = 4

18 What is at the ?

$20 \div 4 = 5$, step 5 · ? = 10

19 What is at the ?

$20 \div 2 = 10$, step 10 · ? = 10

20 The scale is -15, -10, □, 0. What is □?

-15 to -10 is 5, step 5 · □ = -5

Step 5: Halfway between

E6 (WE DO) -4 and 2

6 apart, half is 3 · $-4 + 3 = -1$

21 -6 and 2

8 apart, half is 4 · $-6 + 4 = -2$

22 -10 and 0

10 apart, half is 5 · $-10 + 5 = -5$

23 -3 and 9

12 apart, half is 6 · $-3 + 6 = 3$

24 -11 and -1

10 apart, half is 5 · $-11 + 5 = -6$

25 -15 and 5

20 apart, half is 10 · $-15 + 10 = -5$

Step 6: Number line problems

E7 (WE DO) An ant starts at 3. It walks 7 left. Where is it now?

start at 3, move 7 left · land on -4

26 A frog starts at -4. It hops 9 right. Where is it now?

start at -4, move 9 right · land on 5

27 Point A is at -6. Point B is 10 to the right of A. Where is B?

start at -6, move 10 right · land on 4

28 An ant at 5 walks 8 left, then 2 right. Where is it now?

start at 5, move 8 left: -3 · move 2 right · land on -1

29 A lift at level -3 goes up 7 floors, then down 6. Which level is it on?

start at -3, move 7 up: 4 · move 6 down · land on -2

1.03 Comparing and ordering integers

(booklet p. 7)

Step 1: Which is larger?

E1 (WE DO) -3 or 2

larger: 2

1 -6 or -2

larger: -2

2 1 or -4

larger: 1

3 -8 or -3

larger: -3

4 -1 or -7

larger: -1

5 -12 or -9
larger: -9

Step 2: Using $<$ and $>$

E2 (WE DO) $-4 \square 3$
larger: 3 · symbol: $<$

6 $-2 \square -6$
larger: -2 · symbol: $>$

7 $0 \square -5$
larger: 0 · symbol: $>$

8 $-9 \square -3$
larger: -3 · symbol: $<$

9 $4 \square -10$
larger: 4 · symbol: $>$

10 $-15 \square -14$
larger: -14 · symbol: $<$

11 $-20 \square 1$
larger: 1 · symbol: $<$

Step 3: Ordering

E3 (WE DO) Ascending: $3, -4, 0, -1$
smallest: -4 · $-4, -1, 0, 3$

E4 (WE DO) Descending: $-2, 5, -7, 1$
largest: 5 · $5, 1, -2, -7$

12 Ascending: $2, -3, -1, 4$
smallest: -3 · $-3, -1, 2, 4$

13 Descending: $-4, 6, -1, 0$
largest: 6 · $6, 0, -1, -4$

14 Ascending: $-12, 8, -15, -3, 1$
smallest: -15 · $-15, -12, -3, 1, 8$

15 Descending: $-20, -2, -11, 5, -8$
largest: 5 · $5, -2, -8, -11, -20$

Step 4: Comparing temperatures

E5 (WE DO) Which is colder: -4°C or -9°C ?
 -9 is left of -4 · colder: -9°C

16 Which is colder: -2°C or -6°C ?
 -6 is left of -2 · colder: -6°C

17 Which is colder: 3°C or -1°C ?
 -1 is left of 3 · colder: -1°C

18 Which is warmer: -10°C or -7°C ?
 -7 is right of -10 · warmer: -7°C

19 Which is colder: -21°C or -12°C ?
 -21 is left of -12 · colder: -21°C

Step 5: Simplify, then compare

E6 (WE DO) $-(-3) \square -4$
 $= 3 \square -4$ · symbol: $>$

20 $-(-1) \square -2$
 $= 1 \square -2$ · symbol: $>$

21 $-(+4) \square -3$
 $= -4 \square -3$ · symbol: $<$

22 $-(-5) \square 6$
 $= 5 \square 6$ · symbol: $<$

23 $-7 \square -(+9)$
 $= -7 \square -9$ · symbol: $>$

24 $-(+6) \square 5$
 $= -6 \square 5$ · symbol: $<$

25 $-8 \square -(-8)$
 $= -8 \square 8$ · symbol: $<$

Step 6: Is it right?

E7 (WE DO) Tom says -9 is larger than -4 . Is he right?
 -9 is left of -4 · so -9 is smaller · No. -4 is larger.

26 Mia says $-1 > -10$. Is she right?
 -1 is right of -10 · so -1 is larger · Yes. Mia is right.

27 Ben says -20°C is colder than -2°C . Is he right?
 -20 is left of -2 · so -20°C is colder · Yes. Ben is right.

28 Ali says $-5, -7, 0, 2$ is in ascending order. Is he right?
smallest: -7 · $-7, -5, 0, 2$ · No. It is $-7, -5, 0, 2$.

29 Jo says $-(-6) < -6$. Is Jo right?
 $-(-6) = 6$ · 6 is right of -6 · No. $-(-6) > -6$.

1.04 Adding integers (booklet p. 9)

Step 1: Adding a positive

E1 (WE DO) $-2 + 4$
 $= 2$

1 $-4 + 3$
 $= -1$

2 $-5 + 2$
 $= -3$

3 $-6 + 4$
 $= -2$

4 $-3 + 7$
 $= 4$

5 $-7 + 7$
 $= 0$

Step 2: Adding a negative

E2 (WE DO) $6 + (-2)$
 $= 6 - 2$ · $= 4$

E3 (WE DO) $-2 + (-3)$
 $= -2 - 3$ · $= -5$

6 $7 + (-3)$
 $= 7 - 3$ · $= 4$

7 $5 + (-5)$
 $= 5 - 5$ · $= 0$

8 $2 + (-6)$
 $= 2 - 6$ · $= -4$

9 $4 + (-10)$
 $= 4 - 10$ · $= -6$

10 $-1 + (-4)$
 $= -1 - 4$ · $= -5$

$$11 \quad -3 + (-6) \\ = -3 - 6 \quad \cdot \quad = -9$$

Step 3: Zero pairs

$$E4 \text{ (WE DO)} \quad -5 + 8 \\ 8 - 5 = 3 \quad \cdot \quad + \text{ has more: } 3$$

$$E5 \text{ (WE DO)} \quad 6 + (-9) \\ 9 - 6 = 3 \quad \cdot \quad - \text{ has more: } -3$$

$$12 \quad -7 + 10 \\ 10 - 7 = 3 \quad \cdot \quad + \text{ has more: } 3$$

$$13 \quad -9 + 4 \\ 9 - 4 = 5 \quad \cdot \quad - \text{ has more: } -5$$

$$14 \quad 8 + (-12) \\ 12 - 8 = 4 \quad \cdot \quad - \text{ has more: } -4$$

$$15 \quad -15 + 20 \\ 20 - 15 = 5 \quad \cdot \quad + \text{ has more: } 5$$

$$16 \quad 25 + (-40) \\ 40 - 25 = 15 \quad \cdot \quad - \text{ has more: } -15$$

$$17 \quad -36 + 50 \\ 50 - 36 = 14 \quad \cdot \quad + \text{ has more: } 14$$

Step 4: Three numbers

$$E6 \text{ (WE DO)} \quad 1 + (-4) + 5 \\ = -3 + 5 \quad \cdot \quad = 2$$

$$18 \quad 2 + (-5) + 4 \\ = -3 + 4 \quad \cdot \quad = 1$$

$$19 \quad -3 + 6 + (-2) \\ = 3 + (-2) \quad \cdot \quad = 1$$

$$20 \quad 12 + (-15) + (-6) \\ = -3 + (-6) \quad \cdot \quad = -9$$

$$21 \quad -20 + 8 + 15 \\ = -12 + 15 \quad \cdot \quad = 3$$

Step 5: Adding in real life

$$E7 \text{ (WE DO)} \quad \text{It is } -5^{\circ}\text{C}. \text{ The temperature rises } 8^{\circ}\text{C}. \text{ What is the temperature now?} \\ -5 + 8 \quad \cdot \quad = 3^{\circ}\text{C}$$

$$22 \quad \text{It is } -4^{\circ}\text{C} \text{ at dawn. By lunch it has warmed up } 9^{\circ}\text{C}. \text{ What is the temperature at lunch?} \\ -4 + 9 \quad \cdot \quad = 5^{\circ}\text{C}$$

$$23 \quad \text{A lift is at level } -2. \text{ It goes up 5 floors. Which level is it on now?} \\ -2 + 5 \quad \cdot \quad = \text{level } 3$$

$$24 \quad \text{Sam has } \$15 \text{ in the bank. Sam spends } \$40 \text{ using credit. What is the balance now?} \\ 15 + (-40) \quad \cdot \quad = -\$25$$

$$25 \quad \text{A diver is at } -18 \text{ m. She swims up 7 m, then down 4 m. Where is she now?} \\ -18 + 7 + (-4) \quad \cdot \quad = -11 + (-4) \quad \cdot \quad = -15 \text{ m}$$

Step 6: Is it right?

$$E8 \text{ (WE DO)} \quad \text{Tom says } -6 + 2 = -8. \text{ Is he right?} \\ \text{start at } -6, \text{ move 2 right } \quad \cdot \quad = -4 \quad \cdot \quad \text{No. The answer is } -4.$$

$$26 \quad \text{Mia says } -8 + 5 = -13. \text{ Is she right?} \\ \text{start at } -8, \text{ move 5 right } \quad \cdot \quad = -3 \quad \cdot \quad \text{No. The answer is } -3.$$

$$27 \quad \text{Ben says } -7 + 3 + (-3) = -7. \text{ Is he right?} \\ 3 + (-3) \text{ is a zero pair } \quad \cdot \quad -7 + 0 = -7 \quad \cdot \quad \text{Yes. Ben is right.}$$

$$28 \quad \text{Ali says } -4 + \square = 2 \text{ means } \square = -2. \text{ Is he right?} \\ \text{from } -4 \text{ to } 2 \text{ is 6 right } \quad \cdot \quad \square = 6 \quad \cdot \quad \text{No. } \square \text{ is 6, not } -2.$$

$$29 \quad \text{Jo says adding always makes a number bigger. Is Jo right?} \\ 4 + (-6) = -2 \quad \cdot \quad -2 \text{ is smaller than } 4 \quad \cdot \quad \text{No. It can get smaller.}$$

1.05 Subtracting integers (booklet p. 11)

Step 1: Subtracting a positive

$$E1 \text{ (WE DO)} \quad 1 - 4 \\ = -3$$

$$1 \quad 3 - 5 \\ = -2$$

$$2 \quad 0 - 6 \\ = -6$$

$$3 \quad 2 - 7 \\ = -5$$

$$4 \quad 4 - 12 \\ = -8$$

$$5 \quad 5 - 11 \\ = -6$$

Step 2: Starting below zero

$$E2 \text{ (WE DO)} \quad -2 - 3 \\ \text{move 3 left } \quad \cdot \quad = -5$$

$$6 \quad -1 - 4 \\ \text{move 4 left } \quad \cdot \quad = -5$$

$$7 \quad -5 - 2 \\ \text{move 2 left } \quad \cdot \quad = -7$$

$$8 \quad -6 - 6 \\ \text{move 6 left } \quad \cdot \quad = -12$$

$$9 \quad -10 - 5 \\ \text{move 5 left } \quad \cdot \quad = -15$$

$$10 \quad -8 - 12 \\ \text{move 12 left } \quad \cdot \quad = -20$$

$$11 \quad -25 - 15 \\ \text{move 15 left } \quad \cdot \quad = -40$$

Step 3: Subtracting a negative

$$E3 \text{ (WE DO)} \quad 4 - (-2) \\ = 4 + 2 \quad \cdot \quad = 6$$

$$12 \quad 6 - (-3) \\ = 6 + 3 \quad \cdot \quad = 9$$

$$13 \quad 1 - (-7) \\ = 1 + 7 \quad \cdot \quad = 8$$

$$14 \quad -2 - (-5) \\ = -2 + 5 \quad \cdot \quad = 3$$

$$15 \quad -8 - (-3) \\ = -8 + 3 \cdot = -5$$

$$16 \quad -4 - (-4) \\ = -4 + 4 \cdot = 0$$

$$17 \quad -15 - (-20) \\ = -15 + 20 \cdot = 5$$

Step 4: Mixed + and -

$$E4 \text{ (WE DO)} \quad 2 - (-3) - 6 \\ = 2 + 3 - 6 \cdot = 5 - 6 \cdot = -1$$

$$18 \quad 1 - (-4) - 7 \\ = 1 + 4 - 7 \cdot = 5 - 7 \cdot = -2$$

$$19 \quad -3 - 5 - (-2) \\ = -3 - 5 + 2 \cdot = -8 + 2 \cdot = -6$$

$$20 \quad -6 - (-9) + (-5) \\ = -6 + 9 + (-5) \cdot = 3 + (-5) \cdot = -2$$

$$21 \quad 10 - (-2) - 15 \\ = 10 + 2 - 15 \cdot = 12 - 15 \cdot = -3$$

Step 5: Change and difference

$$E5 \text{ (WE DO)} \quad \text{It was } -3^{\circ}\text{C, then } 5^{\circ}\text{C. How much did it rise?} \\ 5 - (-3) \cdot = 5 + 3 \cdot = 8^{\circ}\text{C}$$

$$22 \quad \text{It was } -6^{\circ}\text{C, then } 2^{\circ}\text{C. How much did it rise?} \\ 2 - (-6) \cdot = 2 + 6 \cdot = 8^{\circ}\text{C}$$

$$23 \quad \text{A gull is at 25 m. A diver is at } -15 \text{ m. How far apart are they?} \\ 25 - (-15) \cdot = 25 + 15 \cdot = 40 \text{ m}$$

$$24 \quad \text{A freezer is } -18^{\circ}\text{C. The kitchen is } 22^{\circ}\text{C. What is the difference?} \\ 22 - (-18) \cdot = 22 + 18 \cdot = 40^{\circ}\text{C}$$

$$25 \quad \text{It was } 4^{\circ}\text{C, then } -9^{\circ}\text{C. What was the change?} \\ -9 - 4 \cdot = -13^{\circ}\text{C}$$

Step 6: Is it right?

$$E6 \text{ (WE DO)} \quad \text{Tom says } -4 - 3 = -1. \text{ Is he right?} \\ \text{start at } -4, \text{ move 3 left } \cdot = -7 \cdot \text{ No. The answer is } -7.$$

$$26 \quad \text{Mia says } 2 - (-6) = -4. \text{ Is she right?} \\ = 2 + 6 \cdot = 8 \cdot \text{ No. The answer is 8.}$$

$$27 \quad \text{Ben says } -5 - (-5) = 0. \text{ Is he right?} \\ = -5 + 5 \cdot = 0 \cdot \text{ Yes. Ben is right.}$$

$$28 \quad \text{Ali says } -1 - 8 = -9. \text{ Is he right?} \\ \text{start at } -1, \text{ move 8 left } \cdot = -9 \cdot \text{ Yes. Ali is right.}$$

$$29 \quad \text{Jo says subtracting always makes a number smaller. Is Jo right?} \\ 4 - (-3) = 7 \cdot 7 \text{ is larger than } 4 \cdot \text{ No. It can get larger.}$$

1.06 Multiplying integers (booklet p. 13)

Step 1: Equal groups

$$E1 \text{ (WE DO)} \quad 2 \times (-4) \\ (-4) + (-4) \cdot = -8$$

$$1 \quad 2 \times (-3) \\ (-3) + (-3) \cdot = -6$$

$$2 \quad 3 \times (-4) \\ (-4) + (-4) + (-4) \cdot = -12$$

$$3 \quad 4 \times (-2) \\ (-2) + (-2) + (-2) + (-2) \cdot = -8$$

$$4 \quad 3 \times (-10) \\ (-10) + (-10) + (-10) \cdot = -30$$

$$5 \quad 2 \times (-25) \\ (-25) + (-25) \cdot = -50$$

Step 2: The sign rules

$$E2 \text{ (WE DO)} \quad -3 \times 4 \\ \text{different signs: } - \cdot = -12$$

$$6 \quad 5 \times (-2) \\ \text{different signs: } - \cdot = -10$$

$$7 \quad -6 \times 3 \\ \text{different signs: } - \cdot = -18$$

$$8 \quad -4 \times (-2) \\ \text{same signs: } + \cdot = 8$$

$$9 \quad 7 \times (-5) \\ \text{different signs: } - \cdot = -35$$

$$10 \quad -9 \times (-3) \\ \text{same signs: } + \cdot = 27$$

$$11 \quad -8 \times 6 \\ \text{different signs: } - \cdot = -48$$

Step 3: Why - x - is +

$$E3 \text{ (WE DO)} \quad 1 \times (-5) = -5, 0 \times (-5) = 0, \dots \\ -1 \times (-5) = 5 \cdot -2 \times (-5) = 10$$

$$12 \quad 1 \times (-4) = -4, 0 \times (-4) = 0, \dots \\ -1 \times (-4) = 4 \cdot -2 \times (-4) = 8$$

$$13 \quad 1 \times (-2) = -2, 0 \times (-2) = 0, \dots \\ -1 \times (-2) = 2 \cdot -2 \times (-2) = 4$$

$$E4 \text{ (WE DO)} \quad -2 \times (-6) \\ \text{same signs: } + \cdot = 12$$

$$14 \quad -5 \times (-4) \\ \text{same signs: } + \cdot = 20$$

$$15 \quad -7 \times (-8) \\ \text{same signs: } + \cdot = 56$$

Step 4: Three numbers

$$E5 \text{ (WE DO)} \quad 2 \times (-3) \times 4 \\ = -6 \times 4 \cdot = -24$$

$$16 \quad -2 \times 5 \times 3 \\ = -10 \times 3 \cdot = -30$$

$$17 \quad 3 \times (-2) \times (-5) \\ = -6 \times (-5) \cdot = 30$$

$$18 \quad -3 \times 4 \times (-5) \\ = -12 \times (-5) \cdot = 60$$

$$19 \quad -4 \times (-2) \times (-3) \\ = 8 \times (-3) \cdot = -24$$

$$20 \quad -5 \times (-2) \times (-2) \times (-3) \\ = 10 \times (-2) \times (-3) \cdot = -20 \times (-3) \cdot = 60$$

Step 5: Multiplying in real life

E6 (WE DO) The temperature drops 2°C every hour for 4 hours. What is the total change?

$$4 \times (-2) \cdot = -8^{\circ}\text{C}$$

21 A diver goes down 4 m every minute for 6 minutes. What is her change in height?

$$6 \times (-4) \cdot = -24 \text{ m}$$

22 Sam spends \$5 a day for 7 days. What is the change in Sam's money?

$$7 \times (-5) \cdot = -\$35$$

23 A share price falls \$3 a week for 9 weeks. What is the total change?

$$9 \times (-3) \cdot = -\$27$$

24 A lift at level 3 goes down 2 floors, 5 times. Which level is it on now?

$$5 \times (-2) = -10 \cdot 3 + (-10) \cdot = \text{level } -7$$

Step 6: Is it right?

E7 (WE DO) Tom says $-3 \times 6 = 18$. Is he right?
different signs: $- \cdot = -18$ · No. The answer is -18 .

25 Mia says $-7 \times (-2) = -14$. Is she right?
same signs: $+ \cdot = 14$ · No. The answer is 14.

26 Ben says $-4 \times \square = 28$ means $\square = -7$. Is he right?
 $-4 \times (-7) \cdot = 28$ · Yes. Ben is right.

27 Ali says $(-5)^2 = -25$. Is he right?
 $= -5 \times (-5) \cdot = 25$ · No. The answer is 25.

28 Jo says $-2 \times (-3) \times (-4)$ is positive. Is Jo right?
3 negatives: odd, so $- \cdot = -24$ · No. It is -24 .

1.07 Dividing integers (booklet p. 15)

Step 1: Division undoes $\times$

$$\text{E1 (WE DO)} \quad 4 \times (-2) = -8 \\ -8 \div 4 = -2 \cdot \quad -8 \div (-2) = 4$$

$$1 \quad 2 \times (-5) = -10 \\ -10 \div 2 = -5 \cdot \quad -10 \div (-5) = 2$$

$$2 \quad 3 \times (-6) = -18 \\ -18 \div 3 = -6 \cdot \quad -18 \div (-6) = 3$$

$$3 \quad 5 \times (-4) = -20 \\ -20 \div 5 = -4 \cdot \quad -20 \div (-4) = 5$$

$$4 \quad 6 \times (-7) = -42 \\ -42 \div 6 = -7 \cdot \quad -42 \div (-7) = 6$$

Step 2: The sign rules

$$\text{E2 (WE DO)} \quad -18 \div 3 \\ \text{different signs: } - \cdot = -6$$

$$5 \quad -12 \div 4 \\ \text{different signs: } - \cdot = -3$$

$$6 \quad 20 \div (-5) \\ \text{different signs: } - \cdot = -4$$

$$7 \quad -30 \div (-6) \\ \text{same signs: } + \cdot = 5$$

$$8 \quad -42 \div 7 \\ \text{different signs: } - \cdot = -6$$

$$9 \quad -56 \div (-8) \\ \text{same signs: } + \cdot = 7$$

$$10 \quad 72 \div (-9) \\ \text{different signs: } - \cdot = -8$$

Step 3: Missing numbers

$$\text{E3 (WE DO)} \quad \square \div 2 = -4 \\ \square = -4 \times 2 \cdot \quad \square = -8$$

$$\text{E4 (WE DO)} \quad -20 \div \square = 5 \\ \square = -20 \div 5 \cdot \quad \square = -4$$

$$11 \quad \square \div 3 = -6 \\ \square = -6 \times 3 \cdot \quad \square = -18$$

$$12 \quad \square \div (-4) = 5 \\ \square = 5 \times (-4) \cdot \quad \square = -20$$

$$13 \quad -30 \div \square = -6 \\ \square = -30 \div (-6) \cdot \quad \square = 5$$

$$14 \quad \square \div (-7) = -3 \\ \square = -3 \times (-7) \cdot \quad \square = 21$$

$$15 \quad 48 \div \square = -8 \\ \square = 48 \div (-8) \cdot \quad \square = -6$$

Step 4: Dividing step by step

$$\text{E5 (WE DO)} \quad -24 \div 2 \div 3 \\ = -12 \div 3 \cdot = -4$$

$$16 \quad -40 \div 4 \div 2 \\ = -10 \div 2 \cdot = -5$$

$$17 \quad 36 \div (-3) \div 2 \\ = -12 \div 2 \cdot = -6$$

$$18 \quad -48 \div (-4) \div (-3) \\ = 12 \div (-3) \cdot = -4$$

$$19 \quad -90 \div 3 \div (-5) \\ = -30 \div (-5) \cdot = 6$$

Step 5: Sharing equally

E6 (WE DO) A debt of \$20 is shared by 4 people. Write each share as an integer.
 $-20 \div 4 \cdot = -\$5 \text{ each}$

20 A debt of \$36 is shared by 3 people. Write each share as an integer.
 $-36 \div 3 \cdot = -\$12 \text{ each}$

21 The temperature fell 15°C in 5 hours. What was the change each hour?
 $-15 \div 5 \cdot = -3^{\circ}\text{C each hour}$

22 A diver went down 24 m in 6 minutes. What was her change each minute?
 $-24 \div 6 \cdot = -4 \text{ m each minute}$

23 A share price fell \$56 over 8 days, the same each day. What was the change each day?
 $-56 \div 8 \cdot = -\$7 \text{ each day}$

Step 6: Is it right?

E7 (WE DO) Tom says $-24 \div 6 = 4$. Is he right?
different signs: $- \cdot = -4$ · No. The answer is -4 .

24 Mia says $35 \div (-7) = -5$. Is she right?
different signs: $- \cdot = -5$ · Yes. Mia is right.

25 Ben says $-40 \div (-8) = -5$. Is he right?
same signs: $+ \cdot = 5$ · No. The answer is 5.

26 Ali says $\square \div (-2) = 6$ means $\square = -3$. Is he right?
 $\square = 6 \times (-2)$ · $\square = -12$ · No. $\square$ is -12 , not -3 .

27 Jo says dividing by a negative always gives a negative.
Is Jo right?
 $-12 \div (-3) = 4$ · 4 is positive · No. It can be positive.

1.08 Order of operations with integers

(booklet p. 17)

Step 1: The order

E1 (WE DO) $-5 + 2 \times 3$
 $= -5 + 6$ · $= 1$

1 $-1 + 2 \times 5$
 $= -1 + 10$ · $= 9$

2 $-10 + 3 \times 2$
 $= -10 + 6$ · $= -4$

3 $6 - 4 \times 3$
 $= 6 - 12$ · $= -6$

4 $-8 + 12 \div 4$
 $= -8 + 3$ · $= -5$

5 $-7 - 10 \div 2$
 $= -7 - 5$ · $= -12$

6 $4 \times (-3) + 5$
 $= -12 + 5$ · $= -7$

Step 2: Brackets first

E2 (WE DO) $(-2 + 5) \times 3$
 $= 3 \times 3$ · $= 9$

7 $(-4 + 6) \times 5$
 $= 2 \times 5$ · $= 10$

8 $(1 - 7) \times 2$
 $= -6 \times 2$ · $= -12$

9 $(-3 - 2) \times 4$
 $= -5 \times 4$ · $= -20$

10 $(-8 + 2) \div 3$
 $= -6 \div 3$ · $= -2$

11 $-2 \times (5 - 9)$
 $= -2 \times (-4)$ · $= 8$

12 $(-9 - 6) \div (-5)$
 $= -15 \div (-5)$ · $= 3$

Step 3: $\times$ and $\div$ before $+$ and $-$

E3 (WE DO) $2 \times (-3) + 12 \div 4$
 $= -6 + 3$ · $= -3$

13 $3 \times (-2) + 8 \div 2$
 $= -6 + 4$ · $= -2$

14 $-12 \div 3 - 2 \times 5$
 $= -4 - 10$ · $= -14$

15 $4 \times (-5) + (-18) \div 6$
 $= -20 + (-3)$ · $= -23$

16 $-3 \times (-4) - 20 \div (-5)$
 $= 12 - (-4)$ · $= 12 + 4$ · $= 16$

Step 4: Left to right

E4 (WE DO) $-20 \div 5 \times 2$
 $= -4 \times 2$ · $= -8$

E5 (WE DO) $3 - 7 + 2$
 $= -4 + 2$ · $= -2$

17 $-18 \div 6 \times 3$
 $= -3 \times 3$ · $= -9$

18 $6 - 10 + 3$
 $= -4 + 3$ · $= -1$

19 $24 \div (-4) \times 2$
 $= -6 \times 2$ · $= -12$

20 $-2 - 5 + 9$
 $= -7 + 9$ · $= 2$

21 $-30 \div (-5) \times (-2)$
 $= 6 \times (-2)$ · $= -12$

Step 5: Putting it together

E6 (WE DO) $5 - (2 - 6) \times 3$
 $= 5 - (-4) \times 3$ · $= 5 - (-12)$ · $= 17$

22 $-4 + (1 - 3) \times 5$
 $= -4 + (-2) \times 5$ · $= -4 + (-10)$ · $= -14$

23 $10 - (-2 + 6) \times 3$
 $= 10 - 4 \times 3$ · $= 10 - 12$ · $= -2$

24 $(-8 + 2) \div 3 - 4$
 $= -6 \div 3 - 4$ · $= -2 - 4$ · $= -6$

25 $-3 \times (4 - 7) + (-10)$
 $= -3 \times (-3) + (-10)$ · $= 9 + (-10)$ · $= -1$

Step 6: Is it right?

E7 (WE DO) Tom says $-2 + 3 \times 4 = 4$. Is he right?
 $3 \times 4 = 12$ first · $-2 + 12 = 10$ · No. The answer is 10.

26 Mia says $6 - 2 \times (-3) = 12$. Is she right?
 $2 \times (-3) = -6$ first · $6 - (-6) = 12$ · Yes. Mia is right.

27 Ben says $-12 \div 2 \times 3 = -2$. Is he right?
left to right: $-12 \div 2 = -6$ · $-6 \times 3 = -18$ · No. The answer is -18 .

28 Ali says $(-5 + 3) \times 2 = -4$. Is he right?
brackets first: -2 · $-2 \times 2 = -4$ · Yes. Ali is right.

29 Jo says $-4 + 6 \times 3 = 6$. Is Jo right?
 6×3 first: $-4 + 18$ · $= 14$ · No. $(-4 + 6) \times 3 = 6$.

1.09 Integers on a calculator (booklet p. 19)

Step 1: The $(-)$ key

E1 (WE DO) $-48 + 25$
 $(-)$ $48 + 25 =$ · $= -23$

$$1 \quad -63 + 41$$

$$(-) 63 + 41 = \cdot = -22$$

$$2 \quad -27 - 38$$

$$(-) 27 - 38 = \cdot = -65$$

$$3 \quad -150 + 86$$

$$(-) 150 + 86 = \cdot = -64$$

$$4 \quad 45 - 112$$

$$45 - 112 = \cdot = -67$$

$$5 \quad -208 - 97$$

$$(-) 208 - 97 = \cdot = -305$$

$$6 \quad -76 + 176$$

$$(-) 76 + 176 = \cdot = 100$$

Step 2: $\times$ and $\div$ with negatives

$$E2 \text{ (WE DO)} -18 \times 12$$

$$(-) 18 \times 12 = \cdot = -216$$

$$7 \quad -23 \times 15$$

$$(-) 23 \times 15 = \cdot = -345$$

$$8 \quad 36 \times (-14)$$

$$36 \times (-) 14 = \cdot = -504$$

$$9 \quad -540 \div 12$$

$$(-) 540 \div 12 = \cdot = -45$$

$$10 \quad -27 \times (-31)$$

$$(-) 27 \times (-) 31 = \cdot = 837$$

$$11 \quad -1\,196 \div (-26)$$

$$(-) 1196 \div (-) 26 = \cdot = 46$$

Step 3: Check the sign first

$$E3 \text{ (WE DO)} -34 \times 29$$

$$\text{different signs: } - \cdot = -986$$

$$12 \quad -52 \times (-18)$$

$$\text{same signs: } + \cdot = 936$$

$$13 \quad 65 \times (-24)$$

$$\text{different signs: } - \cdot = -1\,560$$

$$14 \quad -3\,024 \div 56$$

$$\text{different signs: } - \cdot = -54$$

$$15 \quad -2\,491 \div (-47)$$

$$\text{same signs: } + \cdot = 53$$

$$16 \quad -125 \times 16$$

$$\text{different signs: } - \cdot = -2\,000$$

Step 4: Brackets on the calculator

$$E4 \text{ (WE DO)} (-28 + 13) \times 6$$

$$\text{bracket: } -15 \cdot = -90$$

$$17 \quad (-45 - 19) \times 5$$

$$\text{bracket: } -64 \cdot = -320$$

$$18 \quad (36 - 81) \times (-4)$$

$$\text{bracket: } -45 \cdot = 180$$

$$19 \quad (-120 + 48) \div 8$$

$$\text{bracket: } -72 \cdot = -9$$

$$20 \quad (-67 - 58) \div (-25)$$

$$\text{bracket: } -125 \cdot = 5$$

Step 5: Estimate, then calculate

$$E5 \text{ (WE DO)} -39 \times 21$$

$$\approx -40 \times 20 = -800 \cdot = -819$$

$$21 \quad -52 \times 19$$

$$\approx -50 \times 20 = -1\,000 \cdot = -988$$

$$22 \quad -61 \times (-29)$$

$$\approx -60 \times (-30) = 1\,800 \cdot = 1\,769$$

$$23 \quad 78 \times (-41)$$

$$\approx 80 \times (-40) = -3\,200 \cdot = -3\,198$$

$$24 \quad -403 + 198$$

$$\approx -400 + 200 = -200 \cdot = -205$$

Step 6: Calculator problems

$$E6 \text{ (WE DO)} \text{ An account has \$120. Four payments of \$38 come out. What is the balance?}$$

$$120 - 4 \times 38 \cdot = -\$32$$

$$25 \quad \text{A submarine at } -185 \text{ m rises 12 m a minute for 9 minutes. Where is it now?}$$

$$-185 + 9 \times 12 \cdot = -77 \text{ m}$$

$$26 \quad \text{It is } -14^\circ\text{C. It warms } 3^\circ\text{C an hour for 7 hours. What is the temperature?}$$

$$-14 + 7 \times 3 \cdot = 7^\circ\text{C}$$

$$27 \quad \text{An account has \$85. Seven payments of \$26 come out. What is the balance?}$$

$$85 - 7 \times 26 \cdot = -\$97$$

$$28 \quad \text{A diver at } -42 \text{ m goes down 6 m a minute for 8 minutes. Where is she now?}$$

$$-42 - 8 \times 6 \cdot = -90 \text{ m}$$

1.10 Integer problems (booklet p. 21)

Step 1: Temperature

$$E1 \text{ (WE DO)} -6^\circ\text{C, rises } 4^\circ\text{C}$$

$$-6 + 4 \cdot = -2^\circ\text{C}$$

$$1 \quad -3^\circ\text{C, rises } 9^\circ\text{C}$$

$$-3 + 9 \cdot = 6^\circ\text{C}$$

$$2 \quad 2^\circ\text{C, falls } 5^\circ\text{C}$$

$$2 - 5 \cdot = -3^\circ\text{C}$$

$$3 \quad -1^\circ\text{C, falls } 6^\circ\text{C}$$

$$-1 - 6 \cdot = -7^\circ\text{C}$$

$$4 \quad -8^\circ\text{C, rises } 8^\circ\text{C}$$

$$-8 + 8 \cdot = 0^\circ\text{C}$$

$$5 \quad -10^\circ\text{C, falls } 4^\circ\text{C}$$

$$-10 - 4 \cdot = -14^\circ\text{C}$$

$$6 \quad -12^\circ\text{C, rises } 15^\circ\text{C}$$

$$-12 + 15 \cdot = 3^\circ\text{C}$$

Step 2: Above and below sea level

$$E2 \text{ (WE DO)} \text{ A gull at 20 m and a diver at } -5 \text{ m}$$

$$20 - (-5) \cdot = 25 \text{ m}$$

$$7 \quad \text{A boat at 0 m and a diver at } -12 \text{ m}$$

$$0 - (-12) \cdot = 12 \text{ m}$$

8 A cliff top at 30 m and a diver at -8 m
 $30 - (-8) \cdot = 38 \text{ m}$

9 A plane at 150 m and a submarine at -60 m
 $150 - (-60) \cdot = 210 \text{ m}$

10 A diver at -5 m and a wreck at -22 m
 $-5 - (-22) \cdot = 17 \text{ m}$

11 A bird at 42 m and a shark at -37 m
 $42 - (-37) \cdot = 79 \text{ m}$

Step 3: Lifts and floors

E3 (WE DO) A lift at level -1 goes up 4 floors. Which level is it on?
 $-1 + 4 \cdot = \text{level } 3$

12 A lift at level 2 goes down 5 floors. Which level is it on?
 $2 - 5 \cdot = \text{level } -3$

13 A lift at level -3 goes up 8 floors. Which level is it on?
 $-3 + 8 \cdot = \text{level } 5$

14 A lift at level 5 goes down 9 floors. Which level is it on?
 $5 - 9 \cdot = \text{level } -4$

15 A lift at level -2 goes down 2, then up 7. Which level is it on?
 $-2 - 2 + 7 \cdot = \text{level } 3$

Step 4: Money

E4 (WE DO) The balance is \$30. You pay \$45.
 $30 - 45 \cdot = -\$15$

16 The balance is \$12. You pay \$25.
 $12 - 25 \cdot = -\$13$

17 The balance is -\$20. You pay in \$50.
 $-20 + 50 \cdot = \$30$

18 The balance is -\$15. You pay \$30.
 $-15 - 30 \cdot = -\$45$

19 The balance is \$40. You pay \$25, then \$30.
 $40 - 25 - 30 \cdot = -\$15$

20 The balance is -\$60. You pay in \$35, then \$40.
 $-60 + 35 + 40 \cdot = \$15$

Step 5: Several steps

E5 (WE DO) It is -6°C. It warms 2°C an hour for 5 hours. How warm is it now?
 $2 \times 5 = 10, -6 + 10 \cdot = 4^\circ\text{C}$

21 A diver at -30 m rises 4 m a minute for 6 minutes. Where is she?
 $4 \times 6 = 24, -30 + 24 \cdot = -6 \text{ m}$

22 It is 5°C. It cools 3°C an hour for 4 hours. How cold is it now?
 $4 \times (-3) = -12, 5 + (-12) \cdot = -7^\circ\text{C}$

23 A win scores +3, a loss -2. A team wins 4 and loses 7. What is its score?
 $12 + 7 \times (-2) = 12 + (-14) \cdot = -2$

24 Sam has \$50 and pays \$12 a week for 6 weeks. What is the balance?
 $6 \times 12 = 72, 50 - 72 \cdot = -\22

Step 6: Is it right?

E6 (WE DO) Tom says level -3, up 5 floors, is level -8. Is he right?
 $-3 + 5 \cdot = 2 \cdot \text{No. It is at level } 2.$

25 Mia says 12 m and -9 m are 3 m apart. Is she right?
 $12 - (-9) \cdot = 21 \cdot \text{No. They are } 21 \text{ m apart.}$

26 Ben says -\$20, then \$35 in, leaves \$15. Is he right?
 $-20 + 35 \cdot = 15 \cdot \text{Yes. Ben is right.}$

27 Ali says 3°C, falling 8°C, gives -5°C. Is he right?
 $3 - 8 \cdot = -5 \cdot \text{Yes. Ali is right.}$

28 Jo says a score of -4 is better than -9. Is Jo right?
 $-4 \text{ is right of } -9 \cdot \text{so } -4 \text{ is larger} \cdot \text{Yes. Jo is right.}$